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Experimental study of the influence of nitrogen and oxygen on the solubility of solid carbon dioxide in liquid and vapor methane at low temperature

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ABSTRACT

Undesired solid CO₂ formation is a main issue for the liquefied natural gas (LNG) industry and, more recently, for the bioLNG production. The abundance of CO₂ solubility data in liquid methane may not be sufficient to fully understand the problem when important amount of air gases (O₂ and N₂) are present in the natural gas or biomethane. Scarcity and incompleteness of available solubility data involving nitrogen and oxygen motivated the production of original solid-liquid-vapor equilibrium (SLVE) data for the N₂-CH₄-CO₂ and N₂-O₂-CH₄-CO₂ systems. The solubility limit of CO₂ in both liquid and vapor phases is measured in this work in order to allow the definition of solid formation conditions. A static analytic methodology is used for obtaining (p,T,x,y) data at SLVE in the temperature range from 125 to 146 K. Experimental results obtained in this work show that the addition of nitrogen and oxygen in methane decreases the solubility of CO₂ in the liquid phase, which is not in agreement with the qualitative behavior shown by literature data. A thermodynamic model for the calculation of solid-liquid-vapor equilibrium of the N₂-O₂-CH₄-CO₂ system is also presented in this work and compared with experimental data, providing good agreement.

1. INTRODUCTION

The study of the influence of nitrogen and air content on the production of liquefied methane is of great importance for the development of new technologies in the field of LNG or alternative liquid fuels, such as the bioLNG (or liquefied biomethane). Nitrogen is a classical natural gas impurity, usually treated only after liquefaction in the Nitrogen Rejection Unit. The interest on biomethane as an advanced liquid bio-fuel has grown over the past decade, driven by decarbonisation objectives in recent energy policy. Raw biogas produced by anaerobic digestion of organic substrate (agricultural wastes, water treatment and landfill wastes) is mainly composed of methane and carbon dioxide. In some cases air is present in the raw biogas: in biogas produced in landfills (landfill gas), which is originated by spontaneous degradation of organic substances present in wastes, air is seeped into the biogas pipes during the collection process and can represent up to 20% of the volume fraction. Also in the biogas digesters some air can be injected (up to 8%) in order to reduce the hydrogen sulfide production through bacterial oxidation, the so-called biological desulfurization.

Liquefied biomethane is produced at temperature similar to the Liquefied Natural Gas operating conditions, but at lower pressure, usually lower than 3 MPa. During the liquefaction process, CO₂ solidification may occur from both the vapor and liquid phase. The prediction of solid formation conditions from vapor and liquid phases is thus of essential importance for the design and simulation of the liquefaction process, in order to avoid uncontrolled solid formation. Solid CO₂ solubility in liquid and vapor phases can be calculated by means of appropriate thermodynamic models able to compute the phase equilibrium involving a solid phase. Nevertheless, phase equilibrium measurements are needed for the regression of model parameters and for the validation of the model.

A large number of experimental studies provide the CO₂ solubility in liquid and vapor methane in the range of interest, going from the boiling temperature of LNG/bio LNG at atmospheric pressure (around 110 K) to the boiling temperature at liquefaction pressure (around 180 K), and some studies also provide the solubility of carbon dioxide in nitrogen and oxygen, as shown in Table 1. CO₂ solubility in liquid phase is the CO₂ content in liquid phase at the solid-liquid equilibrium (SLE). SLE data usually provide temperature and composition of the liquid phase, but rarely the pressure. CO₂ solubility in vapor phase is the CO₂ content in vapor phase at the solid-vapor equilibrium (SVE). When solid phase is in equilibrium with coexisting vapor and liquid phase the phase equilibrium is called solid-liquid-vapor equilibrium (SLVE). The phase diagram of ternary N₂-CH₄-CO₂ system involving a solid phase has been only partially determined through experimental data. An analysis of the experimental procedure and comparison with model results obtained at same temperature, pressure and global composition, show some incompleteness in available data: declared SLE data^{1,2}, are actually SLVE data providing only the CO₂ composition in the liquid phase. Furthermore no data for the quaternary mixture N₂-O₂-CH₄-CO₂ are available, to the authors' knowledge.

The qualitative phase diagram in Figure 1 is representative of the behavior of the N₂-CH₄-CO₂ mixture in the range of temperature and pressure of interest and can help understanding the problem of undesired solid CO₂ formation during methane liquefaction. Considering the stable phases for the pure components at the temperature and pressure of the measurements carried out in this work, CO₂ is in solid phase, CH₄ is in liquid phase and N₂ is in vapor phase. A mixture of methane and carbon dioxide thus presents a solid-liquid equilibrium (SLE) where solid CO₂ is solubilized in the liquid phase. In the ternary diagram, there is thus a SLE region when the system is rich in methane and carbon dioxide, but poor in nitrogen. On the contrary, when the

system is rich in nitrogen and carbon dioxide, but poor in methane, the system is in the SVE region. If the amount of CO_2 is small, no solid is formed and the mixture rich in N_2 and CH_4 is thus in vapor or liquid phase, or at vapor-liquid equilibrium conditions. The SLE, SVE and VLE regions delimit the SLVE triangular region, where the composition of each phase is defined by the vertices of the triangle: in the studied case, solid phase is pure CO_2 , vapor phase is the vertex with higher N_2 concentration and the liquid phase is the vertex with higher CH_4 concentration. The composition of the phases at SLVE does not depend on the global composition of the system for a ternary system at given T and p. From the study of the ternary phase diagram, one can observe that for assuring that a multicomponent system containing methane and carbon dioxide does not present the risk of solid formation at given temperature, pressure and global composition, it can be not sufficient to check the solubility of the carbon dioxide in the liquid phase. In Figure 1, point A represents a mixture with CO_2 content lower than the CO_2 solubility limit in liquid phase and poor in nitrogen: no solid phase is formed from this mixture at given T and p. Point B represents a mixture with same CO_2 content but richer in N_2 . Point B falls in the SLVE region and thus a solid phase is present, in equilibrium with a liquid and a vapor phase. It is then evident that, in order to evaluate the potential formation of solid CO_2 in the natural gas or biomethane liquefaction, a precise knowledge of the position of the phase boundaries is mandatory. This consideration is valid for both the $\text{N}_2\text{-CH}_4\text{-CO}_2$ and the $\text{N}_2\text{-O}_2\text{-CH}_4\text{-CO}_2$ systems. For improving the comprehension of these complex multicomponent systems at conditions where a solid phase can appear, original SLVE measurements in the low temperature region are thus produced in this work.

2. EXPERIMENTAL SETUP AND PROCEDURE

2.1 Experimental setup

The apparatus used in this work for obtaining low temperature SLVE measurements for the N_2 - CH_4 - CO_2 mixture and N_2 - O_2 - CH_4 - CO_2 mixture is a slightly modified version of the original apparatus used for low temperature vapor-liquid equilibrium (VLE) measurements²⁰ and solubility data^{17, 21}. Recently the apparatus has been used for determining solid formation conditions²².

A scheme of the apparatus is represented in Figure 2. The equilibrium cell (EC) is placed into a 55 liters Dewar, partially filled with liquid nitrogen (LN2). In this way, the cell is surrounded by a nitrogen vapor atmosphere. The equilibrium cell is surrounded by a brass shell closed with a brass cap. The brass shell is wrapped by an electric resistance regulated by a Fuji electric PXR PID temperature regulator. The cold source (liquid nitrogen) coupled with the electric resistance allows temperature regulation in the cell. The liquid nitrogen is supplied from a 176 liters ranger intended for this purpose. The temperature is measured at the top and at the bottom of the cell by means of two PT100 Ω (at 0°C) temperature probes, in order to estimate the temperature gradient between the top and the bottom of the cell and induced by the position of the cold source. The average temperature between the two probes is calibrated on the saturated curves of pure methane and pure oxygen. The pressure in the EC is measured by two PTX611 Druck pressure transducers, one for nominal maximum pressure up to 16 MPa and the other up to 0.16 MPa (this last can be isolated when working pressure is higher than 0.16 MPa). Temperature and pressure calibration procedures are discussed in the Supporting Information. The cell is equipped with two pneumatic ROLSI® samplers with stainless steel capillaries. The temperature of the part of

the liquid capillary between the cell and the ROLSI® samplers is controlled by a heating cartridge connected to a PID regulator in order to avoid undesired solidification and condensation during the sampling operations. The ROLSI® samplers are connected to the Gas Chromatograph (GC) through a thermostated transfer line. The GC device is a Perichrom PR 2100 type equipped with a Thermal Conductivity Detector (TCD) and a Flame Ionisation Detector (FID) for analyzing the composition. A Porapak Q column (80/100 mesh, 1/8" external diameter, 2 mm internal diameter, length depending on the system to study, made in Silcosteel), is used for separating the mixture components in the gas chromatograph. Operational conditions of the GC are resumed in Table 2. The CO₂ content in fluid phases at low temperature SLVE is expected to be in the range from 20 to 10000 ppm. The Thermal Conductivity Detector (TCD) is not sensitive enough for revealing the lowest CO₂ quantities in such a range. As a consequence, the detection of small amounts of CO₂ is performed by a Flame Ionisation Detector (FID). Since the FID cannot directly detect CO₂, a methanizer is installed for reducing CO₂ into methane, which can be detected by the FID. The methanizer is a U-tube containing nickel catalyst and kept at 300°C for allowing the complete conversion of CO₂ into CH₄. The four gases used in this work are provided by Air Liquide with declared purity of 99.995%, except for oxygen with declared purity of 99.999%.

2.2 Experimental procedure

A static analytic approach is used for determining the SLVE data at low temperature for the N₂-CH₄-CO₂ mixture and N₂-O₂-CH₄-CO₂ mixture. In order to determine temperature, pressure, liquid and vapor phase compositions at SLVE conditions, the adopted procedure is composed of three main steps: the loading of the cell, the cooling program for reaching the desired

temperature where solid is expected to form, the composition analysis of liquid and vapor phases.

Once vacuum is obtained into the equilibrium cell thanks to a vacuum pump, the mixture with desired composition is loaded into the cell at ambient temperature. Each pure component is loaded into the cell separately from a gas cylinder. The cell is then isolated and the magnetic stirrer activated for assuring suitable mixing in the cell. Once temperature and composition in the equilibrium cell become stable (i.e. their variation in time is lower than the experimental uncertainty in measuring these quantities), the global composition of the loaded mixture is determined by GC analysis.

After the loading of the cell, the Dewar is filled with liquid nitrogen at the half of its volume, avoiding the contact between the bottom of the cell and the liquid nitrogen. The temperature and the pressure inside the cell are continuously recorded and the system is cooled down by setting the temperature regulator. A first rapid cooling is performed down to a temperature set 5 K higher than the expected solid formation temperature. When the system stably reaches this temperature, a second slow cooling is executed setting the temperature regulator to 146 K. The cooling rate is set to 0.5 K/h, and once the temperature is reached, the system is kept at stable temperature for around 48h, checking the stability of pressure. At the end of this period, vapor and liquid phases are sampled for the composition analysis. A series of different samples are analyzed in order to assess the repeatability of the composition measurement. At least two series of samples are taken and analyzed at significant time distance for assuring the stability of the composition. Once the procedure for liquid and vapor phase composition measurement is successfully completed, the Dewar is refilled with liquid nitrogen and temperature is slowly decreased setting the temperature regulator to 132 K with same cooling rate (0.5 K/h). The

system is thus kept at stable temperature for around 48h, and composition analysis of vapor and liquid phase is performed following the procedure previously described. After that, the Dewar is refilled and temperature is decreased setting the temperature regulator to 125 K with same cooling rate (0.5 K/h) for another composition analysis. The temperature of the cell is then let increasing up to ambient temperature for evacuating the cell and loading a new mixture. The procedure is repeated for four different mixtures, with compositions reported in Table 3.

3. RESULTS AND DISCUSSION

3.1 Experimental Results

For each of the four different series of measurements, namely four different global composition mixtures, the measured composition (mole fraction) of liquid and vapor phases and estimated uncertainties at each temperature and pressure are reported in Table 4. Expanded uncertainty (coverage factor equal to 2) on temperature is estimated to be 0.2 K and expanded uncertainty (coverage factor equal to 2) on pressure 0.003 MPa. Details on calculations of uncertainties are presented in the Supporting Information. The formation of a solid phase at experimental temperature and pressure can be inferred from the material balance on CO₂:

$$S = z_{CO_2} - y_{CO_2}V - x_{CO_2}L \quad (1)$$

with:

$$V + L + S = 1 \quad (2)$$

where z_{CO_2} is the global CO₂ mole fraction of the loaded mixture (known), y_{CO_2} and x_{CO_2} are the measured CO₂ mole fractions in the vapor and liquid phases and V, L and S the mole fractions of

vapor, liquid and solid respectively. The solid phase is assumed to be pure CO_2 . Being z_{CO_2} significantly greater than x_{CO_2} and y_{CO_2} in the measurements performed in this work (see Table 3 and Table 4), one obtains a value of S greater than 0 when replacing the experimental values in eq. (1) and eq. (2), proving that a solid phase is present. Values of S for each measurement are reported in Section 3 of the Supporting Information.

Figure 3 presents a qualitative comparison among the experimental results obtained in this work for $\text{CO}_2\text{-CH}_4\text{-N}_2$ and $\text{CO}_2\text{-CH}_4\text{-N}_2\text{-O}_2$ and the literature data of CO_2 solubility for the binary mixture $\text{CO}_2\text{-CH}_4$ and $\text{CO}_2\text{-CH}_4\text{-N}_2$ as function of temperature. The comparison is only qualitative because the solubility of solid CO_2 in the liquid phase depends also on the pressure of the measurements and on the mole fraction of nitrogen in the liquid phase. Nevertheless, it is possible to observe that the addition of nitrogen and oxygen in methane decreases the solubility of CO_2 in the liquid phase in the temperature range investigated. The effect can be explained studying the CO_2 solubility in liquid oxygen and nitrogen, which is lower than in liquid methane, according to literature data presented in Tab. 1. In the investigated temperature range, one can conclude that methane mole fraction in liquid phase has a major effect on CO_2 solubility in liquid phase. When some nitrogen and oxygen are added to methane, methane mole fraction in liquid phase decreases and thus the CO_2 solubility decreases too. The quantitative comparison with the other literature data for the mixture $\text{CO}_2\text{-CH}_4\text{-N}_2$ is not possible because of the differences in pressure and nitrogen mole fraction of the measurements.

3.2 Model for SLVE calculation

Obtained experimental values can be compared with data from literature, when available, and with model results. The approach chosen for modeling the SLVE is the isofugacity of the solid former component in liquid ($f_{CO_2}^L$), vapor ($f_{CO_2}^V$) and solid phase ($f_{CO_2}^S$), which can be assumed as pure CO₂.

$$f_{CO_2}^L(T, p, \mathbf{x}) = f_{CO_2}^V(T, p, \mathbf{y}) = f_{CO_2}^S(T, p) \quad (3)$$

Where T is the temperature, p the pressure, $\mathbf{x}$ and $\mathbf{y}$ the vectors of compositions of the liquid and vapor phase respectively.

The fugacity of the solid phase can be calculated from the model developed for the solid CO₂ by Jäger and Span²³. The advantage of this model is that the fugacity of the solid CO₂ can be directly calculated from the EoS, without any derivation from a hypothetical subcooled liquid fugacity.

$$f_{CO_2}^S(T, p) = p^0 \exp \left(\frac{g_{CO_2}^S(T, p) - g_{CO_2}^0(T, p)}{RT} \right) \quad (4)$$

The suffix 0 is for the reference state, taken in this work as ideal gas at $p^0 = 0.101325 \text{ MPa}$ and $T^0 = 150 \text{ K}$. $g_{CO_2}^0$ is thus the ideal gas Gibbs free energy, which can be obtained from the reference quality Eos for pure CO₂, reported in Kunz et al. (2007)²⁴.

It has been chosen to compute the fugacity of the fluid phases from the GERG 2008 Equation of State (GERG 2008 EoS), providing high accuracy in representing phase equilibria and thermodynamic properties of natural gas type mixtures in liquid and vapor phases^{24, 25}. In this work, the models for both the solid and the fluid fugacities have been used with the original parameters available from the literature²³⁻²⁵.

The range of validity of an empirical equation of state is, typically, determined by the available data used for regressing the equation parameters. In particular for the CH₄-CO₂ system, data for the fluid phases are available down to 153 K²⁶ and for the N₂-CO₂ system down to 209 K²⁷. Nevertheless the GERG 2008 multi-parameter EoS is developed in order to behave reasonably far beyond the range of temperature and pressure of existing data²⁴. Still, data for the O₂-CH₄ system are not available and then EoS parameters are not regressed for this system. As a consequence, the model is predictive when dealing with the N₂-O₂-CH₄-CO₂ system. Experimental CO₂ solubility in liquid and vapor phase obtained in this work is tabulated in Table 5, together with solubility calculated from the model at experimental temperature, pressure and global composition. The measured CO₂ solubility in liquid phase is lower than solubility obtained by the model at experimental temperature, pressure and global composition: CO₂ solubility in liquid phase is overestimated by the model, as it is possible to see in Figure 4. From both model and experimental results it is possible to observe that CO₂ solubility in liquid phase slightly decreases with decreasing pressure. From Figure 5 it is possible to observe that, according to experimental values, CO₂ solubility in vapor phase increases when some nitrogen is replaced by oxygen, but the phenomenon is not predicted by the model. Unfortunately, no SVE data are available for the O₂-CO₂ system for comparing the solubility of methane in vapor oxygen or nitrogen.

The thermodynamic model allows pointing out the dependence of CO₂ solubility on the density of the solvent. In Figs. 6, 7a, and 7b, the solubility of each series of measurements has been plotted as a function of the density calculated by the thermodynamic model at the experimental temperature, pressure, and CO₂-free mole fraction of the liquid phase (Fig. 6) and of the vapor phase (Figs. 7a and 7b). Fig. 6 allows inferring that, for each series of measurements (i.e. for

each global composition of the mixture, see Tab. 3), the solubility of CO₂ in the liquid phase decreases when density increases. The trend of solubility of CO₂ with respect to the density of the solvent is very similar for the two series 1 and 2 obtained for the N₂-CH₄-CO₂ mixture (full and empty squares in Fig. 6). Also the two series 3 and 4 obtained for the N₂-O₂-CH₄-CO₂ mixture (full and empty triangles in Fig. 6) present a similar behavior. The solubility of CO₂ with respect to the calculated density of the solvent (CO₂-free) in the vapor phase is plotted in Fig. 7a for mixture N₂-CH₄-CO₂, and in Fig. 7b for mixture N₂-O₂-CH₄-CO₂. Also in this case it is possible to observe a very similar behavior of the solubility of CO₂ with respect to the density of the solvent for the series 1 and 2, Fig. 7a, and 3 and 4, Fig. 7b.

Expanded uncertainties (coverage factor equal to 2) associated to the mole fraction values are reported in Tab. 4. For methane, the uncertainty is always lower than 2% of the mole fraction value in the liquid phase and 4% in the vapor phase. For oxygen it is lower than 30% of the mole fraction value in the liquid phase and 8% in the vapor phase. For carbon dioxide, it is lower than 20% of the mole fraction (ppm) value in the liquid phase and 50% in the vapor phase.

CONCLUSIONS

In this work, low temperature phase behavior of the methane-carbon dioxide system with addition of air (nitrogen and oxygen) is studied thanks to original SLVE experimental measurements and data from literature. The study is important for understanding the solid formation conditions and maximum CO₂ allowable content in the LNG and bioLNG production processes, especially when significant amount of air gases are present. For this purpose, the study of the SLE may not be sufficient because the presence of light component (N₂ and O₂) induce the formation of a vapor phase at conditions where no vapor phase exists for the binary

CH₄-CO₂ system. Furthermore, from experimental results obtained in this work, it is possible to conclude that the addition of nitrogen and oxygen in methane decreases the solubility of CO₂ in the liquid phase, for temperatures between 125 and 145 K. More experimental studies are suggested in order to provide a full description of the system behavior as function of temperature, pressure, and nitrogen and/or oxygen mole fraction.

Experimental measurements are necessary for testing and calibrating thermodynamic models able to provide a complete representation of the phase behavior of complex systems. The model presented in this work, coupling a high accuracy EoS for the fluid phase (GERG 2008) and an EoS specifically developed for solid CO₂, agrees with experimental values, even if it overestimates the CO₂ solubility in liquid phase.

TABLES.

Table 1. Selected experimental data sources for the CO₂ solubility in liquid and vapor phase in the temperature range 110-180 K.

Author	Mixture	Kind of data	N	Info
Shen et al.(2012) ¹	CH ₄ -CO ₂	SLVE	9	Tpx
	N ₂ -CH ₄ -CO ₂	SLVE	27	Tpx _{CO2}
Gao et al. (2012) ²	CH ₄ -CO ₂	SLVE	9	Tx
	N ₂ -CH ₄ -CO ₂	SLVE	31	Tpx _{CO2}
Davis et al. (1962) ³	CH ₄ -CO ₂	SLVE	42	Tp,Tx,Ty
Boyle (1987) ⁴	CH ₄ -CO ₂	SLE	5	Tx
Cheung and Zander (1968) ⁵	CH ₄ -CO ₂	SLE	9	Tx
Streich (1970) ⁶	CH ₄ -CO ₂	SLE	12	Tx
Preston et al. (1971) ⁷	CH ₄ -CO ₂	SLE	2	Tx
Voss (1975) ⁸	CH ₄ -CO ₂	SLE	22	Tx
Pikaar (1959) ⁹	CH ₄ -CO ₂	SVE	103	Tpy
		SLE	21	Tpx
Yakimenko et al. (1975) ¹⁴	N ₂ -CO ₂	SLE	9	Tx
Agrawal and Leverman (1974) ¹⁰	CH ₄ -CO ₂	SVE	41	Tpy
	N ₂ -CH ₄ -CO ₂	SVE	19	Tpy
Le and Trebble (2007) ¹¹	CH ₄ -CO ₂	SVE	55	Tpy
	N ₂ -CH ₄ -CO ₂	SVE	24	Tpy
Xiong et al. (2015) ¹²	CH ₄ -CO ₂	SVE	64	Tpy
	N ₂ -CH ₄ -CO ₂	SVE	77	Tpy
Fedorova (1940) ¹³	N ₂ -CO ₂	SLE	4	Tx
	O ₂ -CO ₂	SLE	5	Tx
Yakimenko et al. (1975) ¹⁴	N ₂ -CO ₂	SLE	9	Tx

Amamchyan et al. (1973) ¹⁵	O ₂ -CO ₂	SLE	7	Tx
Rest et al. (1990) ¹⁶	N ₂ -CO ₂	SLE	5	Tx
	O ₂ -CO ₂	SLE	3	Tx
De Stefani et al. (2002) ¹⁷	O ₂ -CO ₂	SLE	13	Tx
Sonntag and Van Wilen (1961) ¹⁸	N ₂ -CO ₂	SVE	64	Tpy
Smith et al. (1964) ¹⁹	N ₂ -CO ₂	SVE	72	Tpy

Table 2. Gas Chromatography operating conditions used in this work

Mixture	N ₂ -CH ₄ -CO ₂	N ₂ - O ₂ -CH ₄ -CO ₂
Column	4m Porapak Q	2x4m Porapak Q
Oven Temperature	343 K	248 K for 6.5 min 343 K for 14 min
N ₂ retention time (min)	0.9	5.3
O ₂ retention time (min)	-	6.3
CH ₄ retention time (min)	1.25	10.2
CO ₂ retention time (min)	2.5	16
Carrier gas	He	He
Carrier flow (ml/min)	21	21
TCD temperature	393 K	393 K
FID temperature	573 K	573 K
H ₂ flow in the methanizer	30 ml/min	30 ml/min

Table 3. Global composition of loaded mixture for each series of measurements. N is the series of measurements.

N	N ₂ mol/mol	O ₂ mol/mol	CH ₄ mol/mol	CO ₂ mol/mol
1	0.4	-	0.58	0.02
2	0.19	-	0.79	0.02
3	0.31	0.09	0.58	0.02
4	0.15	0.04	0.79	0.02

Table 4. SLVE experimental liquid (x) and vapor (y) compositions measured in this work for the mixture of N₂ (1)-O₂ (2)-CH₄ (3)-CO₂ (4) and corresponding expanded uncertainties U (coverage factor equal to 2). N is the series of measurements, corresponding to the global composition reported Table 3

N	T (K)	P (MPa)	x ₂	U(x ₂)	x ₃	U(x ₃)	x ₄ (ppm)	U(x ₄) (ppm)	y ₂	U(y ₂)	y ₃	U(y ₃)	y ₄ (ppm)	U(y ₄) (ppm)
1.1	145.9	2.04	-	-	0.763	0.008	6909	900	-	-	0.387	0.012	470	26
1.2	132.1	1.36	-	-	0.734	0.010	1632	331	-	-	0.264	0.009	48	15
1.3	124.9	0.77	-	-	0.705	0.010	749	59	-	-	0.205	0.008	16	8
2.1	145.8	1.12	-	-	0.885	0.005	5722	736	-	-	0.558	0.011	572	22
2.2	132.1	0.77	-	-	0.902	0.004	1185	148	-	-	0.470	0.012	51	12
2.3	124.5	0.52	-	-	0.919	0.004	532	39	-	-	0.444	0.012	15	5
3.1	146.1	2.11	0.070	0.004	0.743	0.005	6948	78	0.119	0.006	0.385	0.006	1542	28
3.2	132.4	1.35	0.083	0.006	0.704	0.012	1709	48	0.129	0.004	0.268	0.004	209	4
3.3	125	0.96	0.084	0.006	0.708	0.007	367	27	0.133	0.005	0.227	0.004	46	9
4.1	146.5	1.39	0.027	0.006	0.891	0.007	6547	217	0.072	0.006	0.599	0.013	1392	41
4.2	132.5	0.84	0.032	0.010	0.884	0.013	1544	90	0.088	0.004	0.468	0.006	185	5
4.3	125.1	0.58	0.033	0.005	0.880	0.006	356	27	0.091	0.003	0.414	0.003	35	2

Table 5. Measured and calculated CO₂ solubility in liquid (x_{CO2}) and vapor phase (y_{CO2}) and deviation between measured and calculated values.

N	T	P	x _{CO2,EXP}	x _{CO2,CALC}	Δx _{CO2}	y _{CO2,EXP}	y _{CO2,CALC}	Δy _{CO2}
	(K)	(MPa)	(ppm)	(ppm)	(ppm)	(ppm)	(ppm)	(ppm)
1.1	145.9	2.04	6909	7846	937	470	633	163
1.2	132.1	1.36	1632	2853	1221	48	92	44
1.3	124.9	0.77	749	1573	824	16	28	12
2.1	145.8	1.12	5722	7674	1952	572	689	117
2.2	132.1	0.77	1185	2669	1484	51	120	69
2.3	124.5	0.52	532	1349	817	15	40	25
3.1	146.1	2.11	6948	7831	883	1542	631	-911
3.2	132.4	1.35	1709	2901	1192	209	96	-113
3.3	125	0.96	367	1551	1184	46	30	-16
4.1	146.5	1.39	6547	7986	1439	1392	750	-642
4.2	132.5	0.84	1544	2787	1243	185	121	-64
4.3	125.1	0.58	356	1451	1095	35	41	6

ASSOCIATED CONTENT

Supporting Information.

Additional details on the calibration procedures for temperature, pressure and CG analysis;
uncertainties calculation for temperature, pressure and composition

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ABBREVIATIONS

LNG, liquefied natural gas; bioLNG, liquefied biomethane; SLE, solid-liquid equilibrium; SVE, solid-vapor equilibrium; VLE, vapor-liquid equilibrium; SLVE, solid-liquid-vapor equilibrium; EoS, equation of state; GC, Gas Chromatography; LN2, liquid nitrogen.

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FIGURES

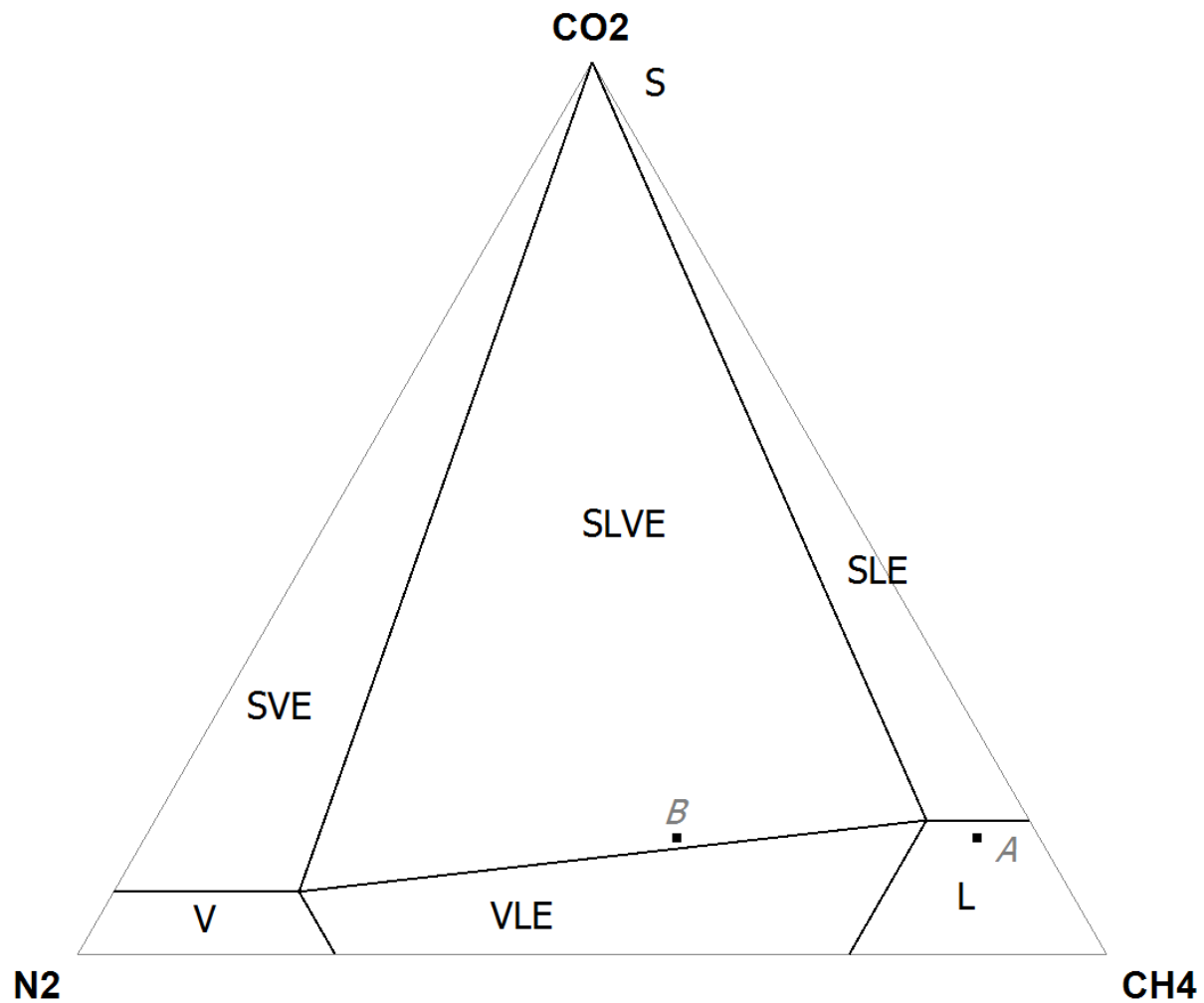


Figure 1. Qualitative phase diagram of the N₂-CH₄-CO₂ system at biomethane liquefaction conditions. S is for solid, L for liquid, and V for vapor phase.

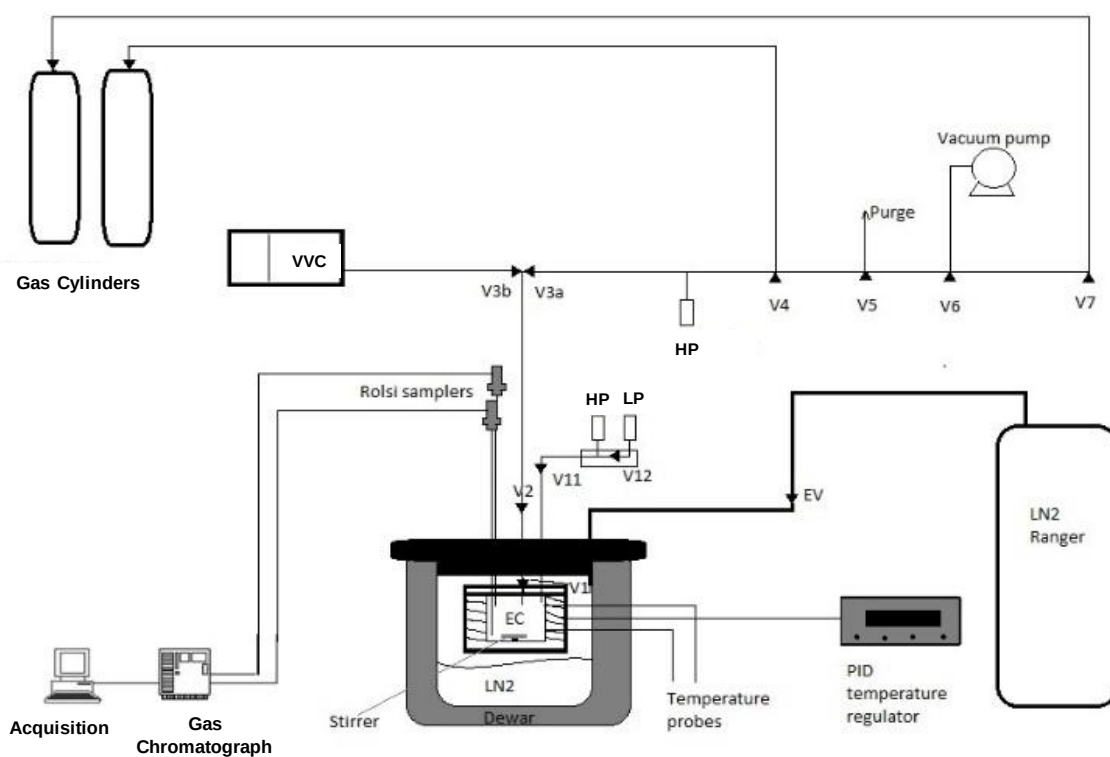


Figure 2. Scheme for the experimental apparatus used in this work. LP: pressure transducer for the low pressure range; EC: equilibrium cell; EV: electrovalve; HP: pressure transducer for the high pressure range; LN2: liquid nitrogen; VVC: variable volume cell; Vi: valves.

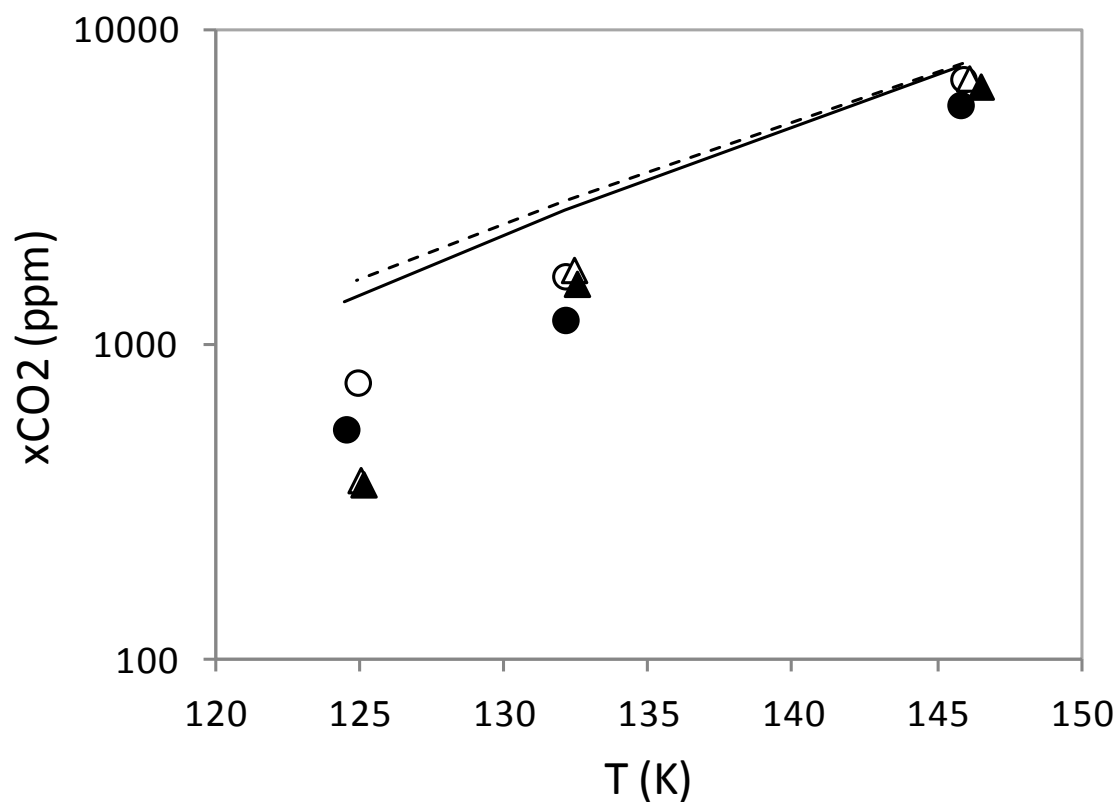


Figure 4. CO₂ solubility in liquid phase (x_{CO_2}) at SLVE conditions from experimental values obtained in this work for the N₂-CH₄-CO₂ system (○ series 1, ● series 2), and the N₂-O₂-CH₄-CO₂ system (△ series 3, ▲ series 4). Results from the model are almost superposed for the ternary and the quaternary system and then they are represented for the ternary system only (--- first series, — second series), for the sake of clarity.

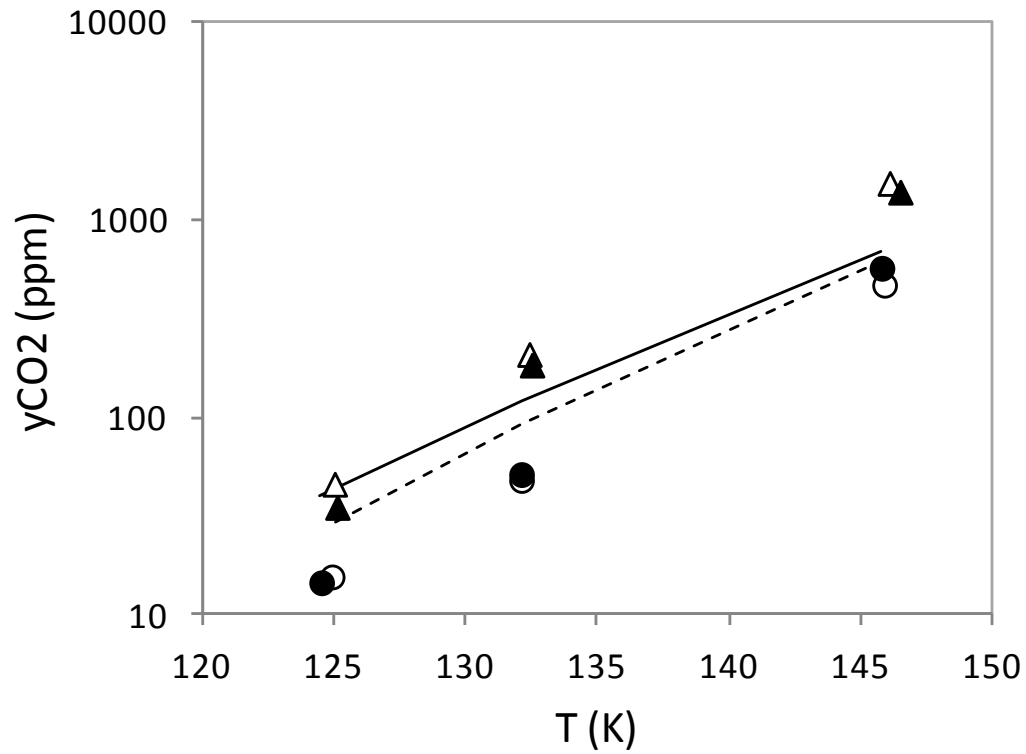


Figure 5. CO₂ solubility in vapor phase (y_{CO_2}) at SLVE conditions from experimental values obtained in this work for the N₂-CH₄-CO₂ system (○ series 3, ● series 4), and the N₂-O₂-CH₄-CO₂ system (△ series 3, ▲ series 4). Results from the model are almost superposed for the ternary and the quaternary system and then they are represented for the ternary system only (--- first series, — second series), for the sake of clarity.

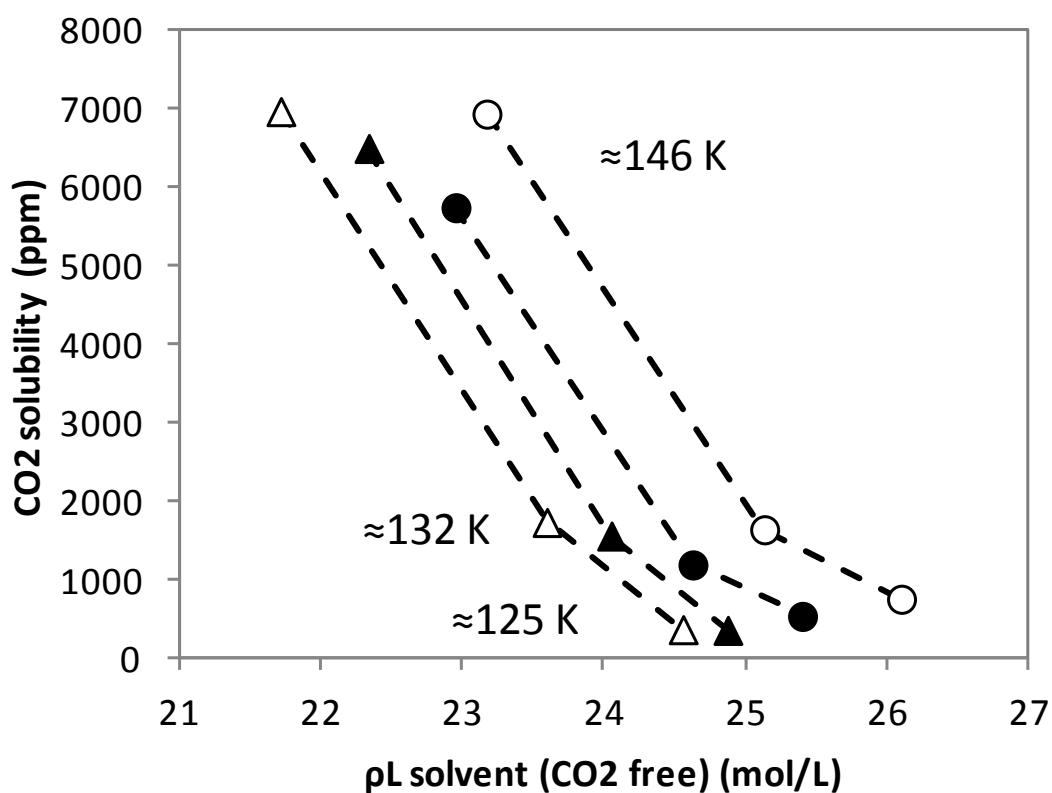


Figure 6. CO₂ solubility in liquid phase with respect to the solvent density (CO₂-free). System N₂-CH₄-CO₂: (○) series 1; (●) series 2. System N₂-O₂-CH₄-CO₂: (△) series 3; (▲) series 4. Density is calculated by the model presented in section 3.2. Dotted lines have the only purpose to connect the symbols in the same series in order to highlight the trends.

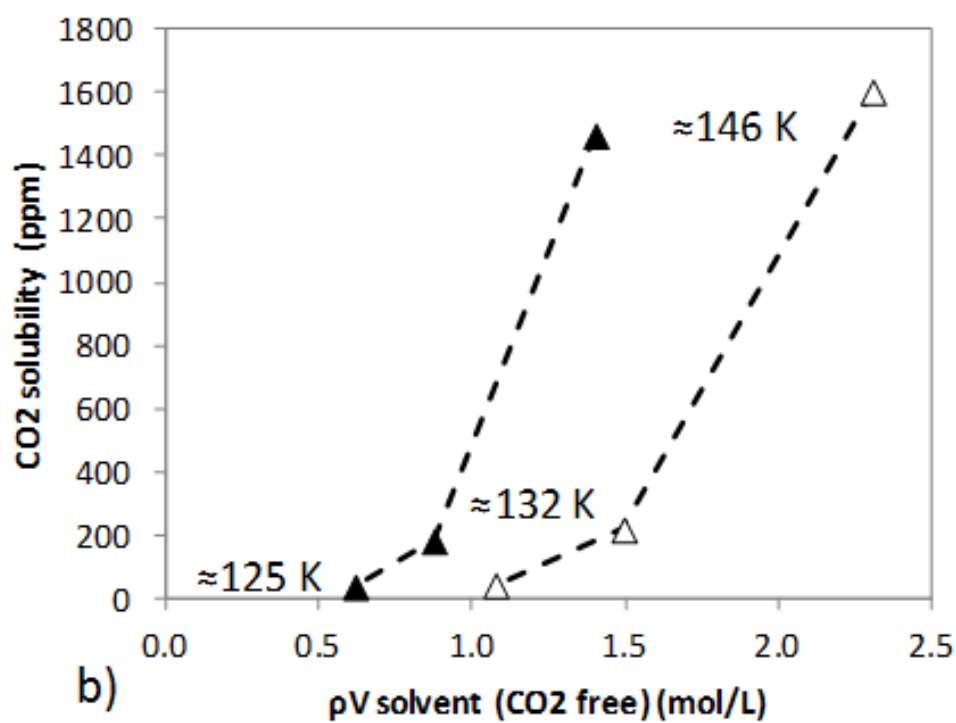
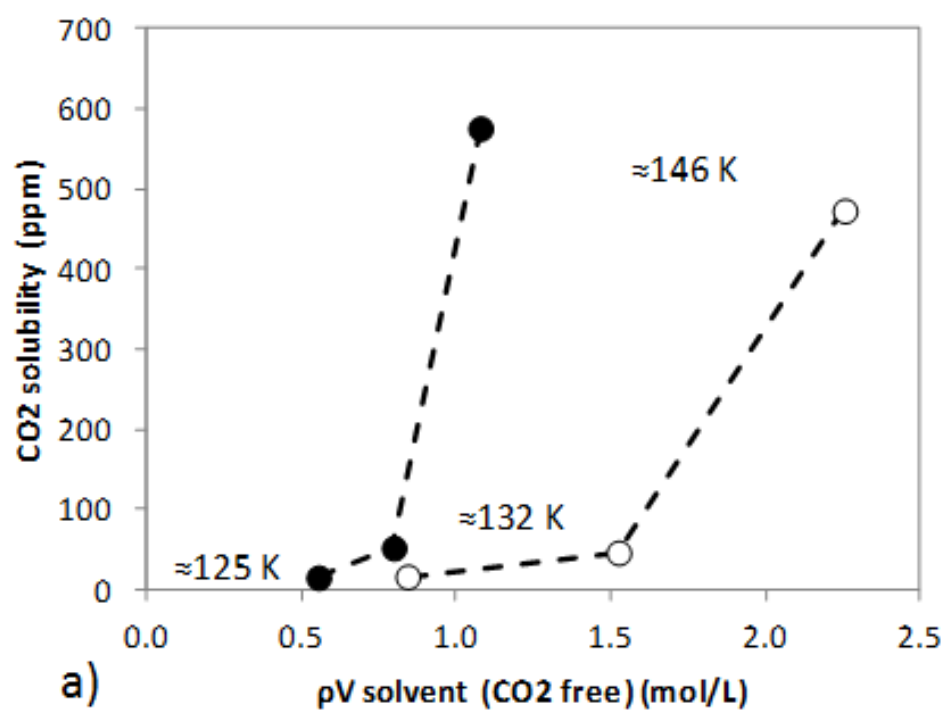


Figure 7. CO₂ solubility in vapor phase with respect to the solvent density (CO₂-free). a) System N₂-CH₄-CO₂: (○) series 1; (●) series 2. b) System N₂-O₂-CH₄-CO₂: (Δ) series 3; (▲) series 4. Density is calculated by the model presented in section 3.2. Dotted lines have the only purpose to connect the symbols in the same series in order to highlight the trends.